What Changes When an LLM Classifies Socratic Math Hints?

Hint transitions across domains, Quarfoot problem types, and a look at my own textbook

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Tea-time talk

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Roadmap

The dataset: SocraticMath

Where it comes from

- Competition problems from the MATH benchmark, each augmented with a *chain of Socratic questions* (Ding et al., CIKM 2024: LLM-generated conversational math lessons)
- One JSON file per problem: `problem`, `level` (1–5), `type`, `solution`, `socratic_questions`

Scale

- 7 domains $\times$ 565–590 problems $=$ **4,043 problems**
- **31,613 Socratic questions** ($\approx$ 7.8 per problem)

domain	problems
algebra	576
counting & probability	584
geometry	575
intermediate algebra	565
number theory	583
prealgebra	590
precalculus	570
total	4,043

Taxonomy 1: four hint types

Goldin, Koedinger & Aleven (2013) — plus one extra type for Socratic chains:

[f] feature-pointing draws attention to a specific aspect of *this* problem
"Notice that these two angles are vertical angles"

[r] principle-stating supplies a general rule, definition, or theorem
"Vertical angles are equal in measure"

[b] bottom-out directly advances the solution — set up, substitute, solve
"Add these two known angles to find the missing one"

[e] extension pushes past the current problem — applications, generalizations
"Can you think of other situations where this matters?"

Goal: learn a 4×4 **transition matrix** M per domain, where p_{ij} is the probability the next hint is type j given the current one is type i.

A worked example (algebra `hint_0`)

Let $f(x) = \begin{cases} ax+3 & x>2 \\ x-5 & -2 \le x \le 2 \\ 2x-b & x<-2 \end{cases}$ Find $a+b$ if f is continuous.

1. [f] Can you explain what a piecewise function is and how it is represented algebraically?
2. [r] How would you determine if the function is continuous where different pieces meet?
3. [f] What conditions need to be satisfied for a piecewise function to be continuous?
4. [b] Can you identify the value of a and b that would make the function continuous?
5. [b] How would you approach solving for a and b?
6. [b] Are there any restrictions or constraints on the values of a and b?
7. [e] Can you explain the significance of continuity in real-life applications?
8. [e] Can you think of other examples where continuity is important?

Labels shown are my hand labels from the report. On this chain Claude reproduces all 8/8 (the regex rubric got 7/8, reading #3 as bottom-out).

Taxonomy 2: Quarfoot's nine problem types

A musical metaphor, ordered (roughly) least $\rightarrow$ most mathematical substance:

	type	what it is
FN	First Notes	one-step, single-path, algorithmic; fluency
AC	Accidentals	looks routine, hides a twist; targets a misconception
CH	Chords	one idea, several steps; deepens a single concept
ET	Etudes	drills one technique; word-problem dressing
IM	Improvisations	a playground; first attempts fail, so you experiment
IN	Interpretations	wide-open solution space; compare qualitatively different paths
FP	First Pieces	first problems needing multiple ideas plus a macro-level plan
SP	Showpieces	very hard, hinges on a flash of insight; cognitive
MP	Masterpieces	many skills converging on a generalizable truth; capstones

The nine types, illustrated from my number theory textbook

Claude's classification of *Olympiad Number Theory* problems (533 total), one example each:

type	problem	source
FN	Find x such that $1 + x + x^2 + x^3 + \cdots = 4$.	Mandelbrot
AC	Show there are no integers x, y with $1691x + 1349y = 1$.	Ch. 5
CH	The sum of 18 consecutive positive integers is a perfect square. Find the smallest possible such sum.	AMC 12
ET	Compute $\sum_{m=1}^{75}(2 + 3m)$.	Ch. 1
IM	Pages numbered 1–n; one page number was accidentally added twice, giving 1986. Which page?	AIME
IN	Prove these Fibonacci identities *in two different ways*.	Ch. 2
FP	Find the 8th term of $1440, 1716, 1848, \ldots$, formed by multiplying corresponding terms of two arithmetic sequences.	AIME
SP	Evaluate $\prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1}$.	Putnam
MP	Prove every positive integer is a sum of non-consecutive Fibonacci numbers.	Zeckendorf

Method: one Claude call per problem, two classifications

- `claude-opus-4-8` receives the problem, difficulty level, worked solution, and the numbered Socratic chain, and returns strict JSON:
  1. one hint label (`f`/`r`/`b`/`e`) per question, in order
  2. one Quarfoot type for the whole problem
- Prompt contains both taxonomies' definitions + the worked example above as the single few-shot anchor
- Every response is **validated** (label count = question count, legal codes only) and retried on failure — batch retry, then per-problem retry
- Downstream analysis is *unchanged* from the regex pipeline: Dirichlet–multinomial posterior per group, row-averaged KL, similarity e^{-KL}
- Everything mirrors the old layout: `classifications_llms/`, `outputs_llms/`

Infrastructure: written by Claude, run on Claude

Built with Claude Code

- Agent wrote the backend, classifier, CLI commands, and plots inside the existing `uv` project
- Backend auto-selects: Anthropic SDK if an API key is set, else headless `claude -p` calls on my Claude subscription
- Batched 10 problems/call ($\approx$ 410 calls), 12 parallel workers

Robustness (needed it!)

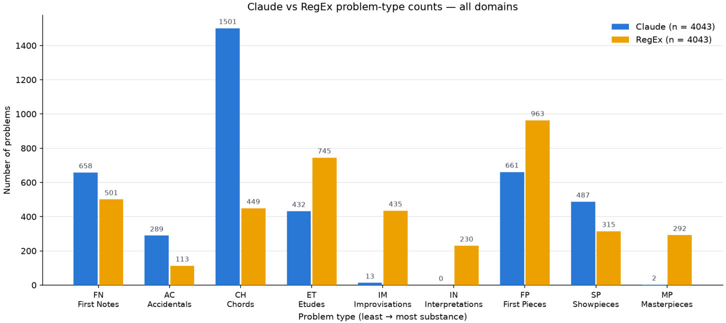
- Resumable: each problem's JSON written on arrival; re-runs skip finished work
- Circuit breaker: 3 consecutive all-fail batches $\Rightarrow$ stop (usage limit), auto-resume later

Compute & cost

- Throughput $\approx$ 53 problems/min $\Rightarrow$ $\approx$ **80 min of active compute** for all 4,043 problems ($\approx$ 3 h wall clock: hit the subscription usage limit twice, auto-resumed)
- Marginal cost: **$0** on the subscription ($\approx$ $60 API-equivalent at Opus list prices)

Claude and the regex rubric disagree — a lot

- Hint labels (MATH): agreement on only **65.0%** of 31,613 questions
- Quarfoot types (MATH): **30.9%** of 4,043 problems
- Quarfoot types (my book): **25.3%** of 533 problems — consistent story



Claude concentrates problems in **Chords** (1,501 vs. 449) and nearly abandons the open-solution-space types: IM 13, IN 0, MP 2 (regex: 435 / 230 / 292).

How one similarity entry is computed (Claude labels)

Count transitions, add the all-ones Dirichlet prior, normalize rows — e.g. the counting & probability f-row has counts $(491, 147, 495, 40)$, so $p_{ff} = \frac{491+1}{1173+4} \approx 0.418$. Doing this for both domains:

$$M_{\text{algebra}} = \begin{pmatrix} 0.227 & 0.224 & 0.498 & 0.051 \\ 0.160 & 0.272 & 0.502 & 0.067 \\ 0.089 & 0.158 & 0.600 & 0.153 \\ 0.030 & 0.073 & 0.193 & 0.704 \end{pmatrix} \quad M_{\text{count. \& prob.}} = \begin{pmatrix} \mathbf{0.418} & 0.126 & 0.421 & 0.035 \\ 0.184 & 0.242 & 0.441 & 0.133 \\ 0.070 & 0.104 & 0.715 & 0.111 \\ 0.030 & 0.060 & 0.079 & \mathbf{0.831} \end{pmatrix}$$

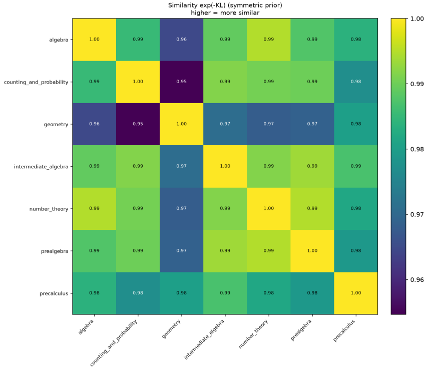
Row-by-row symmetrized KL, then average over the four rows:

row	f	r	b	e	mean
$\frac{1}{2}[KL(A\|C) + KL(C\|A)]$	**0.096**	0.030	0.030	0.063	**0.0549**

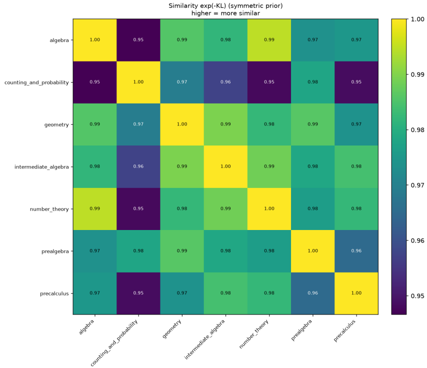
Similarity $= e^{-0.0549} \approx 0.947$ — the *lowest* of any pair under Claude labels. The f-row does the most work: c&p chains feature-pointing $\rightarrow$ feature-pointing at almost $\mathbf{2}\times$ algebra's rate (0.418 vs. 0.227).

The outlier domain flips: geometry → counting & probability

RegEx labels



Claude labels



- RegEx: geometry most dissimilar from everything (the report's conclusion)
- Claude: geometry is among the *most central* domains (mean sim. 0.981); **counting & probability** is the outlier (mean sim. 0.958)

Why counting & probability reads differently

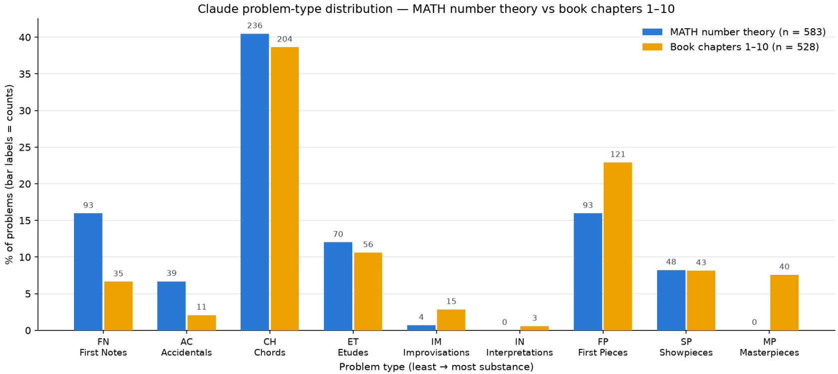
- Its hint *marginals* already stand out — most feature-pointing, least principle-stating of any domain:

	f	r	b	e
counting & probability	**23.9%**	**13.9%**	50.0%	12.1%
all domains (mean)	18.9%	18.6%	51.5%	10.9%

- And its *dynamics*: hints stack observations ($f \to f$ at 0.418) before computing, and once a chain reaches extensions it stays there ($e \to e$ at 0.831 vs. 0.704 for algebra)
- Intuition: "notice how many ways..." beats "recall the theorem..." — combinatorics hints scaffold by *observation*, not by *principle*

Takeaway: with better labels, the geometry-outlier result looks like an artifact of the keyword rubric.

Same classifier, two corpora: MATH number theory vs. book ch. 1–10



- The *middles* match: Chords 40.5% vs. 38.6%; Etudes and Showpieces within ∼1.5 points
- The *tails* diverge in opposite directions

Reading the tails as pedagogy

MATH number theory ($n = 583$)

- Thick floor of fluency checks: FN + AC = **22.6%** ("what is the units digit of…")
- Zero Masterpieces (2 in all 4,043 MATH problems)
- A competition *item bank*

Book chapters 1–10 ($n = 528$)

- Light on drills: FN + AC = 8.7%
- More multi-idea planning: FP 22.9% vs. 16.0%
- 7.6% Masterpieces — culminating capstones (Zeckendorf, USAMO, Putnam)
- A curated *learning arc*

The arc is visible *within* the book too: the share of high-substance problems (FP/SP/MP) climbs from **24%** in Ch. 1 to ≈ **50%** by Ch. 9–10.

Why this is credible: identical classifier, prompt, and rubric on both corpora — near-identical middles, sharply different tails ⇒ properties of the corpora, not classifier noise.

What this offers people who write textbooks

- **An audit tool for your problem mix.** Type profiles make the shape of a book quantitative: drills vs. concept-deepeners vs. capstones, per chapter and overall — and comparable against item banks or other texts.
- **A check on the arc.** You can verify the substance axis actually climbs the way you intended (mine does: 24% $\rightarrow$ 50% high-substance).
- **Gap-finding.** What I learned about my own book:
  - essentially no *Interpretations* (3 of 533) — almost nowhere do I ask students to solve one problem *two different ways* and compare
  - Improvisations are rare too (15) — few low-stakes "playground" problems
  - the capstones I hoped were there really do register (40 Masterpieces)
- **For tutoring systems:** hint scaffolding should adapt per domain — combinatorics wants observation-chaining, algebra wants principle-then-compute.

Takeaways, caveats, next steps

Takeaways

1. LLM labels change the science: the domain outlier flips from geometry to counting & probability, with an interpretable mechanism
2. Problem-type profiles act as corpus fingerprints: item bank vs. curated arc
3. The whole pipeline is cheap enough to be a routine tool ($\approx$ 80 min, $0 marginal)

Caveats

- No human-labeled evaluation set yet — the earlier "gold set" was itself LLM-authored, so I dropped it; a real one is the top priority
- Claude saw the worked solution: single-solution write-ups may bias it against open-ended types (IM/IN) and Masterpieces
- One model, one prompt — no inter-rater κ yet

Next steps: human gold set + Cohen's κ with a second annotator; hint transitions vs. difficulty level; HMM over noisy labels; blind the classifier to the solution text as a robustness check.

Thanks! Questions?

Code available here.